

Arnav Tripathi

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SUMMARY

AI/ML and backend-focused software engineer with hands-on experience building scalable FastAPI services, real-time data pipelines, and generative AI applications. Proven ability to design production-grade backend architecture, integrate third-party APIs, and deliver reliable systems in fast-paced, collaborative environments.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript

Frameworks: FastAPI, React 19, SQLAlchemy, Recharts, Scikit-learn, TensorFlow/Keras, Pandas, NumPy

Backend & Systems: PostgreSQL, WebSockets, OAuth 2.0, REST APIs, Webhooks, APScheduler, Neo4j

AI/ML: Random Forest, LSTM, Computer Vision

Tools: VS Code, Git/GitHub, Figma — **Cloud:** AWS

Data & Analytics: SQL/MySQL, Data Analysis, Machine Learning, Generative AI, Prompt Engineering, Statistical Modelling

PROJECTS

Smart Traffic Violation Reporting System [link](#)

2024

- Secured Second Runner-up at SOLVIT Hackathon competing against 150+ teams
- Built a system using advanced image recognition to detect violations (speeding, red-light running, no seat belt) in real time
- Implemented automated evidence capture (images/video) with instant alert dispatch to law enforcement agencies

Pathwise [link](#)

Nov 2025 – Mar 2026

- Built an interactive React 19 + Vite UI with Framer Motion animations and Recharts for data visualization
- Integrated AI-generated course content with Markdown/LaTeX rendering, quizzes, and gamification features
- Implemented PDF/ZIP export functionality and OpenRouter AI integration for dynamic learning experiences

Project Aura: AI Chronic Disease Management

Mar 2026

- Developed an end-to-end AI platform for chronic disease management, integrating hybrid models (Random Forest & LSTM) for predictive health analytics
- Built a scalable FastAPI backend to handle real-time dataset ingestion and high-concurrency model inference requests
- Implemented specialized data preprocessing pipelines for multi-modal health records to ensure high prediction sensitivity

EXPERIENCE

Software Development Intern – Backend Engineering

May 2026 – Present

SwarnShruti Technologies India Pvt. Ltd.

Remote

- Engineered a centralized FastAPI backend for Orchestra, a multi-platform developer activity aggregator, using serverless PostgreSQL and SQLAlchemy ORM
- Built real-time webhook ingestion pipelines (GitHub, Discord, Figma) with HMAC-SHA256 security and an async WebSocket-driven state machine for live task updates
- Implemented multi-provider OAuth (GitHub, Discord, Google) and secure backend proxy routes to protect API keys, enabling AI-powered roadmap generation and Neo4j graph syncing
- Collaborated with a 5-member distributed team to resolve schema mismatches and maintain zero-downtime integration

Virtual Intern – Google Cloud Generative AI

Jul 2025 – Sep 2025

NASSCOM

Remote

- Worked on Generative AI concepts including LLMs, GANs, and multimodal AI systems.
- Built AI-based solutions through guided projects and mentorship from Google Cloud experts.
- Gained hands-on exposure to prompt engineering, statistical modelling, and cloud-based AI deployment pipelines.

EDUCATION

VIT Bhopal University

Bhopal, India

Bachelor's Degree, Computer Science – CGPA: 8.23

2023 – 2027

Macrovision Academy

Burhanpur, India

CBSE 12th Board – 74.2%

2023

Bhandari Public School

Khandwa, India

CBSE 10th Board – 92%

2021

CERTIFICATIONS

- NPTEL – Introduction to Machine Learning – IIT Madras (63% Score)
- NPTEL – Marketing Analytics – IIT Kharagpur
- Google – Technical IT Support Certificate
- Oracle – Certified Generative AI Professional
- Coursera – Bits and Bytes of Computer Networking
- VITYarthi – AIML, Programming in Java, Python, and C++